

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1-INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25 100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	B. Ravaja	Reg.No.:	192473008
Department:	B.E - CSE CDS	Date:	

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
2. Cloud Server Memory Bottleneck
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
3. Autonomous Vehicle Data Processing
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
4. AI Inference Edge Device
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
5. High-Performance Gaming Console
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I B. Panwar (19247000) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Jany
Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)	
Q1	1	1.25	1	0.25	0.5	4.5
Q2	0.5	1.25	1	0.25	0.5	4
Q3	0.5	1	1.25	0.5	0.5	3.75
Q4	1	1	1.25	0.5	0.25	4.5
Q5	1	1.25	1	0.25	0.25	3.75

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
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20.5

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Student Name:	B.Poorvaja	Reg.No.:	192473008
Department:	B.E-CSE(DS)	Date:	

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

SMARTPHONE PERFORMANCE OPTIMIZATION

PROBLEM UNDERSTANDING

A smartphone manufacturer observes lag while switching between applications. Application switching requires the processor to access instructions, application data, operating-system data and previously used information. If the required information is not available in the faster cache levels, the processor has to access slower memory, increasing latency.

The solution is to design an efficient hierarchical memory system using L1, L2, L3 cache and DRAM so that frequently accessed data remains close to the processor.

ROLE OF DIFFERENT MEMORY LEVELS

L1 Cache:

L1 is the smallest and fastest cache and is located closest to the CPU core. It stores frequently accessed instructions and data. Its very low latency allows the processor to continue execution without waiting for DRAM.

L2 Cache:

L2 is larger than L1 but slightly slower. When required data is not found in L1, the processor checks L2. It therefore reduces the number of accesses to slower memory.

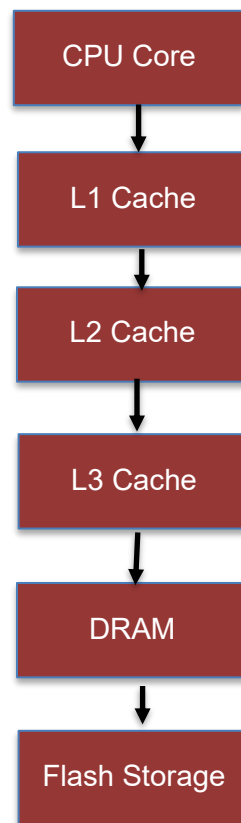
L3 Cache:

L3 is larger than L1 and L2 and is generally shared among processor cores. It stores commonly used data and reduces the frequency of DRAM accesses.

DRAM:

DRAM provides much larger capacity than cache and stores active applications and their working data. However, it has considerably higher latency than cache.

PROPOSED HIERARCHICAL DESIGN



The most frequently accessed instructions and data should be kept in L1 and L2. Shared application data can be retained in L3, while larger application datasets can remain in DRAM.

PROPOSED IMPROVEMENTS

1. Use appropriately sized L1 and L2 caches for each processor core.
2. Increase the effectiveness of L3 cache to reduce DRAM accesses.
3. Use high-speed LPDDR5/LPDDR5X DRAM.
4. Implement hardware prefetching to predict and load frequently required data.
5. Use efficient cache replacement policies to retain useful application data.

6. Keep recently used applications in DRAM so that they do not need to be completely reloaded.

PERFORMANCE ANALYSIS

The average memory access time can be represented as:

$$\text{AMAT} = \text{Hit Time} + (\text{Miss Rate} \times \text{Miss Penalty})$$

By reducing cache misses, the number of expensive DRAM accesses decreases. Therefore, application switching becomes faster and CPU utilization and overall throughput improve.

CONCLUSION

A well-designed L1–L2–L3 cache hierarchy combined with high-speed DRAM can significantly reduce memory-access latency. This provides faster application switching, improved responsiveness and better smartphone performance.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

CLOUD SERVER MEMORY BOTTLENECK

PROBLEM UNDERSTANDING

A cloud service provider experiences high latency during database queries. Database applications frequently access indexes, records and query-related data. If these frequently accessed items are available in cache, the CPU can access them quickly. If they are absent from cache, the system must access DRAM. Excessive virtual-memory paging can increase latency even further.

Therefore, both the cache hierarchy and virtual-memory system must be optimized.

CACHE HIERARCHY VS VIRTUAL MEMORY PAGING

Feature	Cache Hierarchy	Virtual Memory Paging
Purpose	Reduces CPU memory-access latency	Provides larger logical memory space
Main component	L1, L2, L3	DRAM + secondary storage
Access speed	Very fast	Much slower when storage is involved
Data unit	Cache line	Memory page

Feature	Cache Hierarchy	Virtual Memory Paging
Main problem	Cache miss	Page fault
Suitable use	Frequently accessed data	Infrequently accessed data

CACHE HIERARCHY

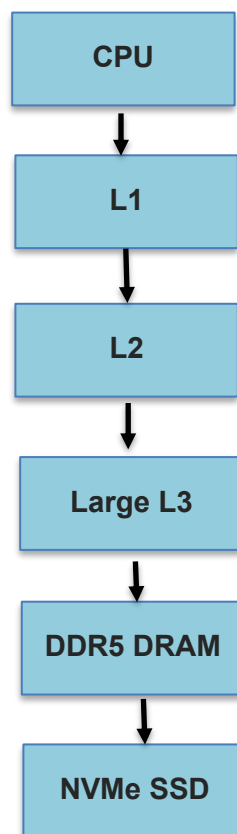
The processor first searches L1, followed by L2 and L3. If the data is not found in the cache hierarchy, it is fetched from DRAM. For a database server, frequently accessed indexes and query data should remain in cache and physical memory as much as possible.

VIRTUAL MEMORY PAGING

Virtual memory allows applications to use a larger logical address space than physical DRAM. When a required page is not present in DRAM, a page fault occurs and the operating system must retrieve the page from secondary storage.

This operation is much slower than accessing DRAM and can significantly increase database query latency.

RECOMMENDED MEMORY DESIGN



IMPROVEMENTS

1. Increase L3 cache capacity where practical.
2. Use high-speed DDR5 DRAM.
3. Provide sufficient DRAM to hold frequently accessed database working sets.
4. Use fast NVMe SSDs for storage and paging when necessary.
5. Optimize database indexes to increase cache and memory locality.
6. Reduce unnecessary page faults by allocating sufficient physical memory.
7. Use prefetching for predictable database access patterns.

PERFORMANCE ANALYSIS

The system should be evaluated using:

- Average database query latency
- Cache hit rate
- Cache miss rate
- Page-fault rate
- DRAM utilization
- Overall query throughput

A higher cache hit rate and lower page-fault rate indicate better memory performance.

CONCLUSION

Cache hierarchy should be used to minimize CPU-access latency, while sufficient DRAM should be provided to reduce virtual-memory paging. Fast NVMe storage can act as secondary storage when paging is unavoidable. This combination reduces database latency and improves cloud-server throughput.

3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

AUTONOMOUS VEHICLE DATA PROCESSING

PROBLEM UNDERSTANDING

An autonomous vehicle continuously receives large amounts of real-time data from sensors such as cameras, LiDAR and radar. This data must be processed quickly to perform tasks such as object detection, obstacle identification and navigation.

A delay in processing sensor data can affect the vehicle's response. Therefore, the memory hierarchy must provide low latency, high bandwidth and sufficient capacity.

COMPARISON OF SRAM, DRAM AND MRAM

Characteristic	SRAM	DRAM	MRAM
Speed	Very high	High	High
Latency	Very low	Higher than SRAM	Low
Density	Low	High	High
Volatility	Volatile	Volatile	Non-volatile
Refresh required	No	Yes	No
Suitable use	Processor cache	Main memory	Fast persistent memory

SRAM

SRAM is extremely fast and has very low latency. It is suitable for storing frequently accessed sensor-processing instructions and critical intermediate data.

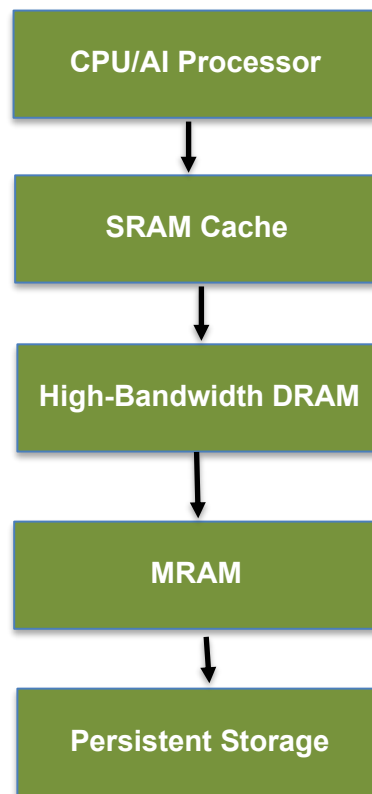
DRAM

DRAM provides significantly higher capacity than SRAM and is therefore suitable for storing large amounts of sensor data, maps and intermediate processing results.

MRAM

MRAM is non-volatile and provides fast access without requiring refresh operations. It can be used for persistent configuration information, frequently required parameters and selected data that should remain available after power loss.

PROPOSED MEMORY HIERARCHY



DESIGN IMPROVEMENTS

1. Use SRAM for the most latency-sensitive processing.
2. Use high-bandwidth DRAM for large sensor datasets.
3. Use multiple DRAM channels to increase memory bandwidth.
4. Use MRAM for important non-volatile information.
5. Use DMA to transfer sensor data directly between devices and memory efficiently.
6. Use prefetching for predictable sensor-data access.
7. Partition memory so that critical real-time tasks receive priority.

PERFORMANCE ANALYSIS

The system should be evaluated using:

- Sensor-data processing latency
- Memory bandwidth
- Cache hit rate
- DRAM utilization

- Data-transfer time
- Real-time response time

The objective is to keep latency low while maintaining sufficient bandwidth for continuous sensor streams.

CONCLUSION

SRAM provides extremely fast access for critical operations, DRAM provides the capacity and bandwidth required for large sensor datasets, and MRAM provides fast non-volatile storage. This hierarchical design allows autonomous vehicles to process sensor information efficiently in real time.

4.AI Inference Edge Device

An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

AI INFERENCE EDGE DEVICE

PROBLEM UNDERSTANDING

An edge AI device needs to load machine-learning models quickly while consuming low power. AI models can contain large numbers of parameters, so the device requires a memory hierarchy that balances speed, capacity, power consumption and persistence.

COMPARISON

Feature	NAND Flash	DRAM	RRAM
Volatility	Non-volatile	Volatile	Non-volatile
Speed	Moderate	High	High
Capacity	High	High	Potentially high
Power	Low standby power	Requires refresh	Low standby power
Main use	Permanent storage	Runtime memory	Fast non-volatile memory

NAND Flash

NAND Flash provides high-capacity, non-volatile storage. Complete AI models, operating-system files and application data can be stored here.

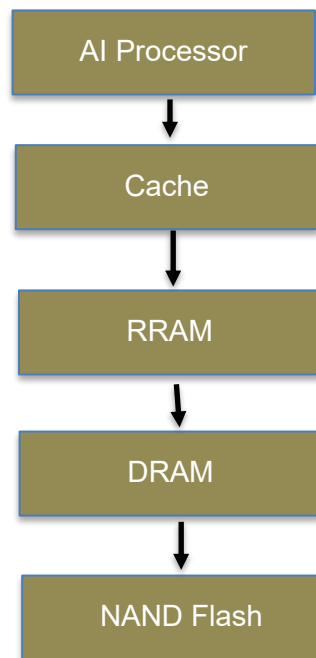
DRAM

DRAM provides fast working memory for AI inference. When a model is executed, the required model parameters and intermediate results can be loaded into DRAM.

RRAM

RRAM is a non-volatile memory technology that can provide fast access with potentially low power consumption. It can be used for frequently accessed AI parameters or selected model data in an optimized architecture.

PROPOSED HIERARCHY



The complete AI model can be permanently stored in NAND Flash. Frequently required data can be moved into DRAM or RRAM for faster access during inference.

IMPROVEMENTS

1. Store complete AI models in NAND Flash.
2. Load only the required model components into DRAM when possible.
3. Use RRAM for frequently accessed parameters where supported.
4. Apply model quantization to reduce model size.
5. Compress model files to reduce loading time.
6. Use efficient data movement between memory levels.
7. Keep frequently used model data in faster memory.

PERFORMANCE ANALYSIS

Important performance parameters include:

- Model loading time
- Inference latency
- Memory bandwidth
- Power consumption
- Memory utilization
- Energy per inference

Reducing unnecessary movement between NAND Flash and DRAM can decrease both loading time and energy consumption.

CONCLUSION

NAND Flash provides large permanent storage, DRAM provides fast runtime memory and RRAM can provide fast non-volatile access. The proposed hierarchy balances model capacity, inference speed and power consumption, making it suitable for edge AI applications.

5.High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

HIGH-PERFORMANCE GAMING CONSOLE

PROBLEM UNDERSTANDING

A gaming console experiences frame drops while loading large scenes. Modern games require continuous access to textures, character models, maps, animations and other assets. If the CPU or GPU has to wait for these assets, frame rendering can be delayed. Therefore, the memory system must provide high bandwidth and low latency.

L3 Cache vs DRAM

Feature	L3 Cache	DRAM
Capacity	Smaller	Much larger
Latency	Very low	Higher
Speed	Very high	High

Feature	L3 Cache	DRAM
Cost per bit	High	Lower
Purpose	Frequently accessed data	Large datasets
Location	Close to processor	Main memory

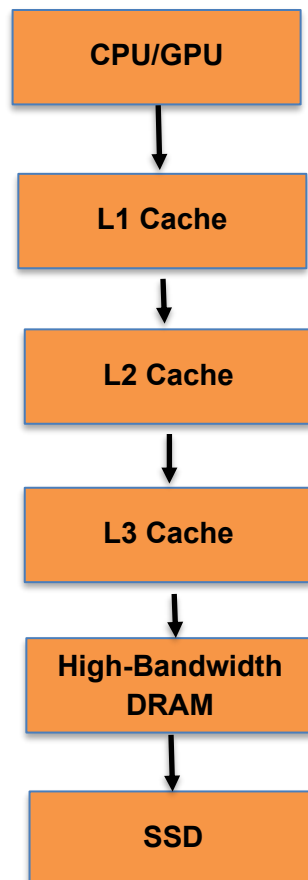
L3 CACHE

L3 cache stores frequently accessed data and reduces the number of accesses to DRAM. Its low latency makes it useful for data that is repeatedly accessed during game execution.

DRAM

DRAM has much greater capacity and can hold large textures, game assets and scene data. However, it has higher access latency than L3 cache.

PROPOSED MEMORY HIERARCHY



Frequently accessed game data should remain in cache, while large scene assets should be stored in high-bandwidth DRAM and loaded from SSD when required.

PROPOSED IMPROVEMENTS

1. Increase effective L3 cache capacity.
2. Use high-bandwidth graphics memory such as GDDR6-class DRAM where appropriate.
3. Use multiple memory channels to increase bandwidth.
4. Implement hardware/software prefetching.
5. Stream game assets from fast SSD storage.
6. Load textures and scene components before they are required.
7. Optimize the game engine to improve memory locality.

PERFORMANCE ANALYSIS

Gaming-memory performance can be evaluated using:

- Memory bandwidth
- Cache hit rate
- Cache miss rate
- Frame rate
- Frame-time consistency
- Scene-loading time
- DRAM utilization

A reduction in cache misses and scene-loading delays reduces CPU/GPU waiting time and helps maintain stable frame rates.

CONCLUSION

L3 cache provides very low-latency access to frequently used game data, while high-bandwidth DRAM provides the capacity needed for large scenes. Combining an effective cache hierarchy, high-bandwidth DRAM and fast SSD storage reduces frame drops and provides smoother gaming performance.

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I _B.Poorvaja(192473008) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

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Signature of the student:

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